An Exploration of How Volume Rendering is Impacted by Lossy Data Reduction

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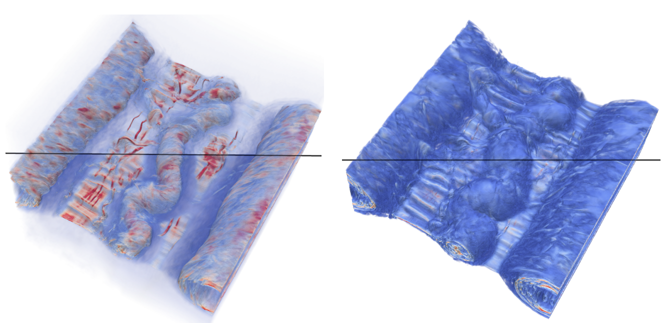
***Abstract*—Data reduction is now frequently used by simulations to reduce the amount of data that needs to be stored. Consequently, several error-bound lossy data reduction techniques have been developed to help compress scientific datasets while trying to maximize quality. However, their impact on visualization has hardly been studied and is not very well understood. In this paper, we do an in-depth analysis of the impact of lossy data reduction on volume rendering to try to determine which parameters, such as characteristics of datasets, opacity, color affect the perception quality of lossy data reduction.**

***Index Terms*—Data Reduction, visualization, volume rendering, study**

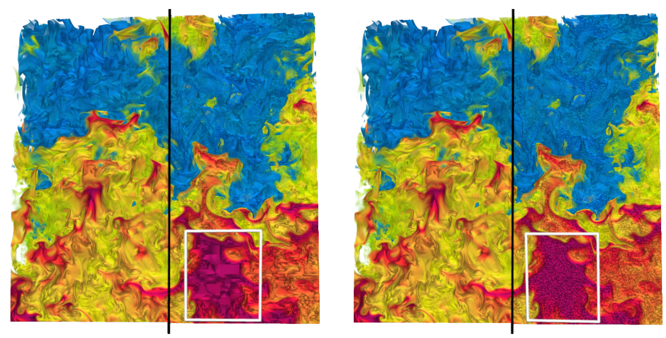
I. Introduction

Simulations are now routinely generating more data than can be stored on High-Performance Computing (HPC) systems. It is common for simulations to use data reduction to decrease the amount of storage required. Since lossless data reduction does not offer enough data reduction, several error-bounded lossy data reduction techniques have been developed, such as SZ3 [2], TTHRESH [3], [4], and SPERR [5], that can significantly decrease the size of simulation data while capping the error using metrics such as absolute error, relative error, and peak signal-to-noise Ratio (PSNR).



(a) Medium opacity, SSIM=0.937 (b) High opacity, SSIM=0.997

Fig. 1: Visualization of the original magnetic reconnection dataset [1] is shown above the black line while visualization of the compressed dataset is below the black line.



(a) Compressor: SZ3 (b) Compressor: TTHRESH

Fig. 2: Visualization of the original Miranda combustion dataset [6] on the left side, and compressed (PSNR value of 47.5 dB) one on the right side of the line.

One of the common uses of scientific data is visualization. Visualization of data is a core tool in scientific discovery and is used for many purposes such as exploration, data verification, image-based analysis, debugging, and communication [7]. Consequently, many data reduction techniques, such as ZFP [8] and Neural Representations, for example, the work of Lu et al. [9] and Qi et al. [10] among others, show visual comparisons of their compressed data as evidence of high-quality compression. However, visualization algorithms are complex, and showing just one image can be misleading. For example, Figure 1 shows a compressed dataset visualized at different opacities. While it is hard to see any difference between the compressed and decompressed image in Figure 1(b), in Figure 1(a), the compressed image (lower half) is "bluer" than the original one (upper half). This is reflected in the SSIM metric, a well-known metric for quantifying perceptual image difference [11], [12], that has a score of 0.997 in Figure 1(b) but only 0.937 in Figure 1(a); an SSIM score of 1 indicates that two images compared are identical.

Lossy data reduction algorithms achieve reductions in size by introducing approximations (or errors) in a dataset. Depending on the type of algorithm used, different types of errors are introduced. Figure 2 shows the same dataset compressed to the same resulting PSNR by SZ3 and TTHRESH. While SZ3 introduces some blocky artifacts, TTHRESH introduces some round circular ones (which can be seen in the white rectangle of Figure 2). Yet, most compressors rely on PSNR to quantify the quality of a dataset. This can be observed by reviewing the publications of common compressors such as SZ [13], ZFP [8], TTHRESH [3], [4], and SPERR [5]. While PSNR is a great metric to quantify error in a dataset, the correct PSNR to produce high-quality visualizations remains unknown. Will it change if different datasets are used in addition to different algorithms? What are other factors we need to consider when predicting the quality of visualizations generated from compressed datasets?

Our main contribution in this paper is an in-depth analysis of how error-bounded lossy data reduction affects visualization created by volume rendering. More specificity, we use:

- three different simulation datasets
- three different compression algorithms
- three different color maps
- three different opacities

to investigate the impact of different amounts of data reduction on visualization. To facilitate analysis, we built a tool (libra) that gathers image perception metrics from different libraries and created a workflow that generates visualization using ParaView [14] from the compressed data. As far as we know, this is the first in-depth study analyzing the factors impacting volume rendering visualization of compressed datasets.

II. Background

A. Data Reduction

Many different data reduction methods exist. In their survey of Error-Bounded Lossy Compression for Scientific Datasets, Di et al. [16] created a taxonomy for lossy data reduction method with the following six classes: i) Decimation/Sampling, ii) Bit Manipulation, iii) Transformation, iv) Prediction, v) Higher-order singular value decomposition (HOSVD), and vi) Deep Learning. As we move from class i) to vi), the quality of the compression tends to increase but so does compression time.

SPERR [5] is a transformation-based compressor that uses wavelets, more specifically it is based on the SPECK coding [17]. SZ3 [2] is a prediction-based compression that uses a dynamic spline interpolation approach to improve the data prediction accuracy, followed by Huffman encoding. TTHRESH [3], [4] is a lossy compressor that utilizes Tucker decomposition (HOSVD) and employs bit-plane coding across the entire set of HOSVD transform coefficients, followed by run-length encoding and arithmetic coding. All of these three compressors are effective at achieving high compression ratios and offer a good rate-distortion trade-off.

B. Color Considerations

Applying color to data is one of the most common and effective ways to convey information about the data. A color map is essentially a transfer function that maps scalar values to color. The choice of color map can dictate what can be seen in the data or modify the relationship between two data points. There is usually a trade-off between the characteristics of a color map and its impact on data [18]. A highly discriminative color map can expose finer detail while a perceptually uniform color map will more accurately relate distances across the data values. A color space is a way to describe the organization and relationship of colors [19]. While there is no color space that aligns completely with human color perception [20], there are commonly used approximations, such as CIELAB and its more recent variant, CIEDE2000. Color spaces using cylindrical coordinates based on hue, saturation, and value (HSV) or hue, saturation, and luminance (HSL) are also considered more perceptually aligned. For purposes of this work, we choose a small number of common color maps with varying properties: the ubiquitous and non-uniform rainbow [21], [22], the classic diverging cool/warm [23], and the perceptually uniform viridis [24].

C. Image Assessment Metrics

Image quality assessment (IQA) is the process of quantifying the accuracy of a modified image with respect to the original. IQA metrics can be either full-reference, meaning the original reference image is available for comparison, or no-reference, where the image quality is assessed solely on the basis of the modified image. Basic pixel-level quantitative IQA metrics such as PSNR and MSE (mean squared error) are good starting points but fail to include human perception of important features in an image. More recent developments in IQA metrics emphasize features of interest and how a human perceives those features in an image. A standard is the structural similarity index (SSIM) metrics [11], [12] which uses local quality measures to emphasize structure. In recent years, a number of Deep Neural Network based metrics have been developed [25] but interestingly, Wang et al. [26] found that they do not out perform the “traditional” IQA metrics for scientific data. Consequently, for purposes of this work, we will use PSNR for assessing dataset quality and SSIM for assessing image quality.

D. Visualization Assessment of Compression

Previous studies have explored how data compression impacts the isosurface visualization of adaptive mesh refinement (AMR) data, highlighting that AMR data’s hierarchical and multi-resolution characteristics introduce unique challenges to its visualization [27]. Furthermore, Wang et al. [26] also explored the impact of volume rendering but they focused on cosmology data and determining which IQA metrics are best suited to capture the changes that we see. In this research, the focus is to understand and explain how the different factors, such as compressor, dataset, opacity, and color impact the visualization of a compressed dataset.

TABLE I: Description of the datasets used for the study.

Dataset (Field)	Domain	Simulation Type	Dataset Statistics	Dimensions / Size
Magnetic Reconnection (Density) [1]	Plasma Physics	3D Particle-In-Cell simulation of magnetic reconnection in plasma	min: 0.0188, max: 24.195, $\overline{x}$: 0.201, σ: 0.483	512 x 512 x 512 512 MB
Miranda (Density) [6]	Fluid Mixing	Large scale-eddy simulation for computing Rayleigh–Taylor instability	min: 0.999, max: 3.00 $\overline{x}$: 1.76, σ: 0.563	1024 x 512 x 1024 2 GB
Nyx (Dark Matter density) [15]	Cosmology	AMR Simulation of baryonic gas coupled with an N-body treatment of the dark matter	min: 0.0188, max:10853.162, $\overline{x}$: 1.0, σ: 7.0	1024 x 1024 x 1024 4 GB

III. Methodology

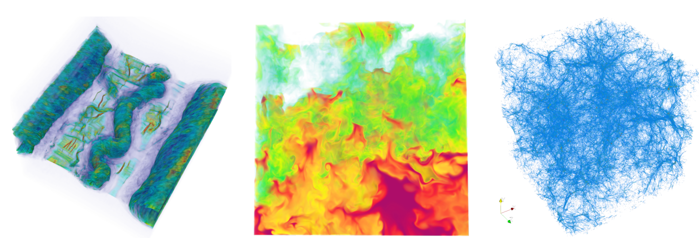
In this study, we investigate how different compression algorithms and parameters affect volume rendering. The variables in the study are: i) data reduction methods, ii) datasets, iii) opacity, and iv) color maps. The parameters are: i) PSNR, ii) Compression Ratio, and iii) Image Quality (SSIM).

A. Experimentation Parameters

1) Data Reduction: For this study, we use three compressors SPERR, SZ3, and TTHRESH from three different domains (as identified by Di et al. [16]): transformation, prediction, and HOSVD classes. Moreover, according to tests run by Li. et al. [5], these three compressors currently provide the highest data reduction quality per compression ratio. One of the questions that this study will try to answer is: *How do different data reduction algorithms affect visualization quality?*

The best way to assess compression quality can be a quite contentious topic. However, PSNR is the most agreed upon metric that has been used by all the leading compressors such as MGARD, SZ, SZ3, SPERR, TTHRESH, and ZFP as well as neural representations in order to demonstrate the performance of their compressor. So, for this paper, we also use PSNR as a base metric to evaluate compressor quality. Our three chosen compressors can all take PSNR as a *minimum* input error-bound. Yet, they do not specify how far above the prescribed value will the resulting PSNR be. So, for our experiment, we use the resulting PSNR value rather than the input one for comparison purposes.

2) Datasets: Different datasets from different simulation domains have very different structures. To make sure that our study is not biased by the dataset that we have chosen, we use three datasets from three different domains as shown in Table I and Figure 3. These datasets have been chosen since they display quite different attributes: the Nyx cosmology dataset has a number of very thin filaments that cover the whole dataset and is roughly isotropic; the Magnetic Reconnection data has a central feature with empty regions above and below; the Miranda dataset covers the whole 3D space with eddies but is anisotropic. This will help us determine how the *characteristics of a dataset impact perception after compression.*



(a) Magnetic Reconnection (b) Miranda (c) Nyx

Fig. 3: Volume rendering of the three test datasets at medium opacity using the rainbow color map.

3) Visualization: Volume Rendering: For this study, we will focus on volume rendering [28], one of the most commonly used visualization techniques for scientific data. The most common model for volume rendering is the emission-absorption model, and the volume rendering integral, which describes how light interacts with a medium, can be expressed as:

$$I(D) = I_0 \cdot e^{-\int_{s_0}^{D} K(t)\,dt} + \int_0^D q(s) \cdot e^{-\int_{s}^{D} K(t)\,dt}\, ds$$

where I_0 is the light entering the volume, $I(D)$ is the light leaving the volume (what we see), $K(t)$ is the absorption coefficient at position t (how much light is absorbed), and q(s) is the emission coefficient at position t. Since we will typically not evaluate the volume rendering equation analytically, the equation is discretized to be used in ray marching algorithm as follows:

$$I(D) = \sum_{i=0}^{n} c_i \prod_{j=i+1}^{n} T_i, \qquad with\ c_o = I(s_0)$$

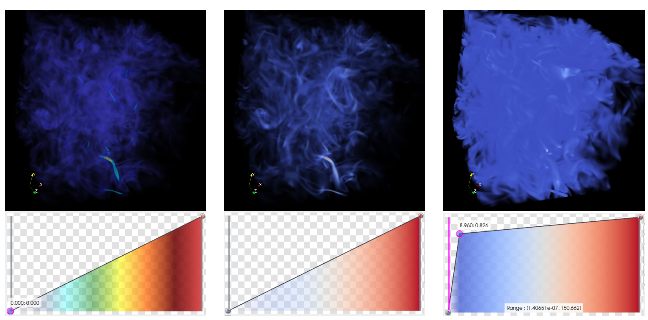
where c_i is the color contribution of for segment i and T_i is the transparency for segment i, along a ray. From a programming perspective, the algorithm can be expressed in pseudocode as shown in Code Listing 1.

Listing 1: Pseudocode for volume rendering

```
-for each pixel
 -cast ray
 -for each voxel intersected
  -get voxel value from dataset
  -look up color and opacity using the
                        transfer function
  - new current pixel color =
     (1-opacity)*pixel_color + alpha*color
  - if opacity threshold == 1
    - stop
```

4) Opacity and Color: As seen in Code Listing 1, the transfer function is what is used in practice to assign colors and opacity. Depending on the color and opacity that is chosen, we can get very different images. Figure 4 shows how visualization can change with different color maps and opacities. This leads us to these two questions: *Does opacity change how compression artifacts are perceived?* and *Does color change how compression artifacts are perceived?*

To answer these questions, we will use three different opacities: low, medium, and high which are characterized as: translucent, a mix of translucent and opaque surfaces, and mostly opaque surfaces respectively as well as three different color maps namely red-blue, rainbow, and viridis. Red-blue is chosen as it is the default in ParaView [14] (a very commonly used scientific visualization package), rainbow has historically been a popular color map and still remains popular in many scientific communities, and viridis is commonly in use by Matplotlib. All the volume-rendered images are generated using ParaView.



Rainbow default transfer function | Red-Blue default transfer function | Red-Blue transfer function with high opacity

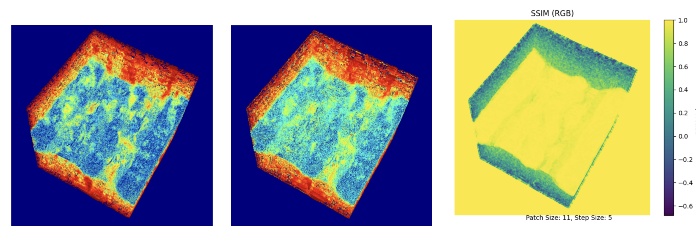
Fig. 4: Isotropic turbulence dataset [29] rendered using different color maps and transfer functions in ParaView.

B. Visualization Quality Assessment

To determine the impact of lossy data reduction on volume rendering, we will use SSIM. SSIM is a full reference metric that is the de facto standard in the image processing community for assessing image quality. In addition to SSIM, visual inspection in different color spaces will be used to confirm the extents of the difference in image quality.

C. Project Libra

To assist us in computing the different metrics and analyzing the results, we have developed Libra, (https://github.com/lanl/libra). Libra is a Python library that leverages image reference metrics from the PyTorch Image Library (piq) [30] and PyTorch Toolbox for Image Quality Assessment (pyiqa) [31], and then allows us to analyze image differences in different color spaces. While computing image differences is common in many tools, they are often in RGB and only provide for absolute differences. Libra allows us to provide a threshold for pixel differences, heat map of color differences and allows us to see color differences in color spaces such as HSV and HSL, which are closer to how humans perceive colors. For example, Figure 5 (a) and (b) show the histogram equalized difference using the RGB and HSV color spaces where we can observe that different color spaces highlight different differences. Also, while SSIM is often given as a single number for a whole image, different regions in an image are impacted differently by compression. Figure 5 (c) shows a map of the different SSIM value differences across the image, clearly showing that it is not uniform. Consequently if the SSIM of an image is low but the Regions of Interests do not seem to be impacted, a "globally" low SSIM image might still be appropriate for scientific usage.



(a) Histogram Equalized difference in the RGB colorspace | (b) Histogram Equalized difference in the HSV colorspace | (c) SSIM map of differences

Fig. 5: Magnetic Reconection dataset compressed using SZ3 with a resulting PSNR value of 65 dB.

IV. Results and Discussion

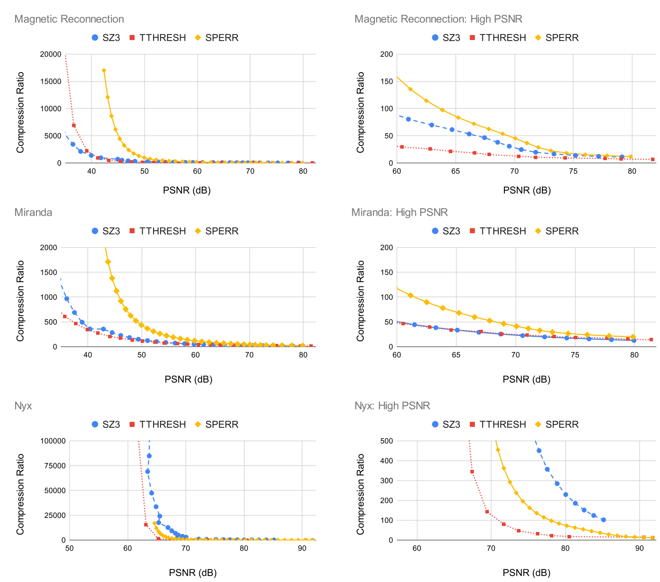


Fig. 6: Compression ranges used during study v/s PSNR of resulting dataset. The right column shows the overall compression, and on the right, the focus is on the higher PSNR values that produce high quality images.

A. Experiment Setup

Our focus in this study is analyzing the impact of error-bounded lossy data reduction on visualization. To do that,

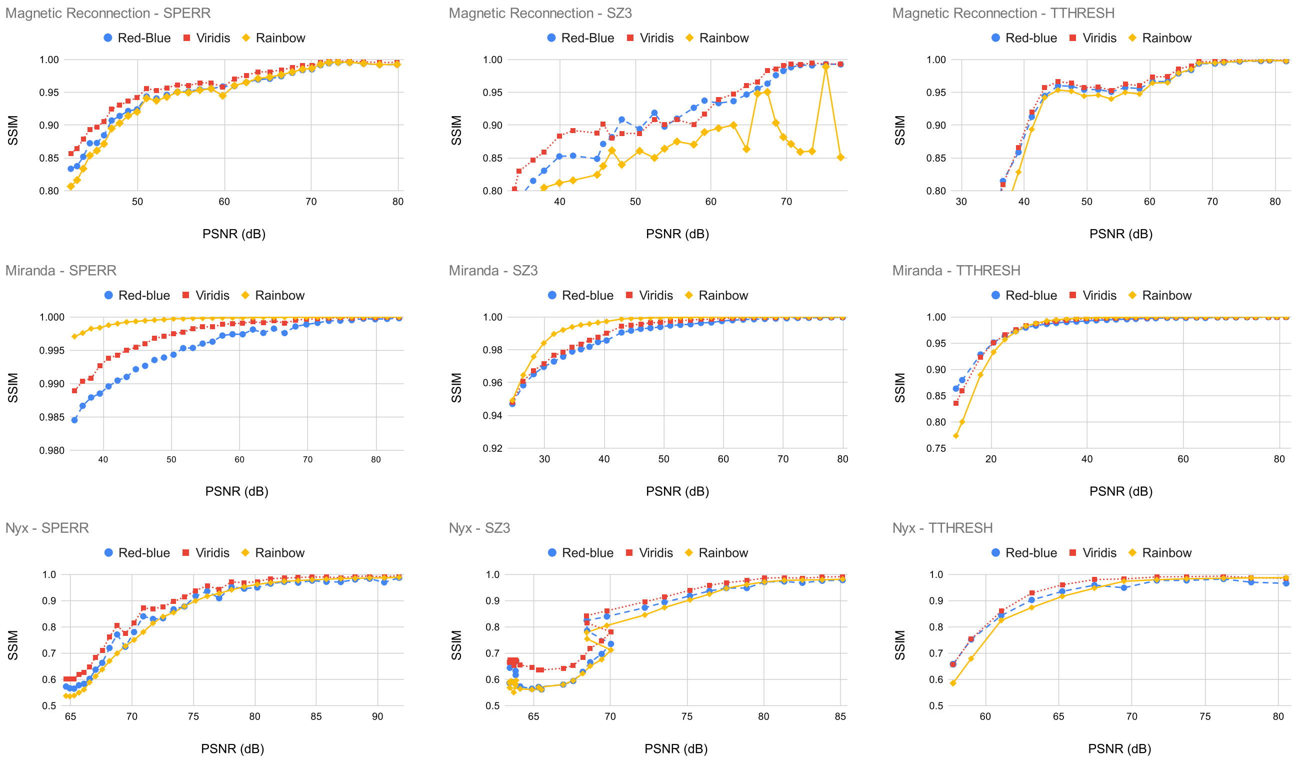


Fig. 7: The title of each graph specifies the dataset and compressor used for volume rendering each of the datasets at medium opacity using at different colormaps. Note that different scales are used for each graph for the vertical axes on the second row, and also that each row of graphs uses different vertical scales.

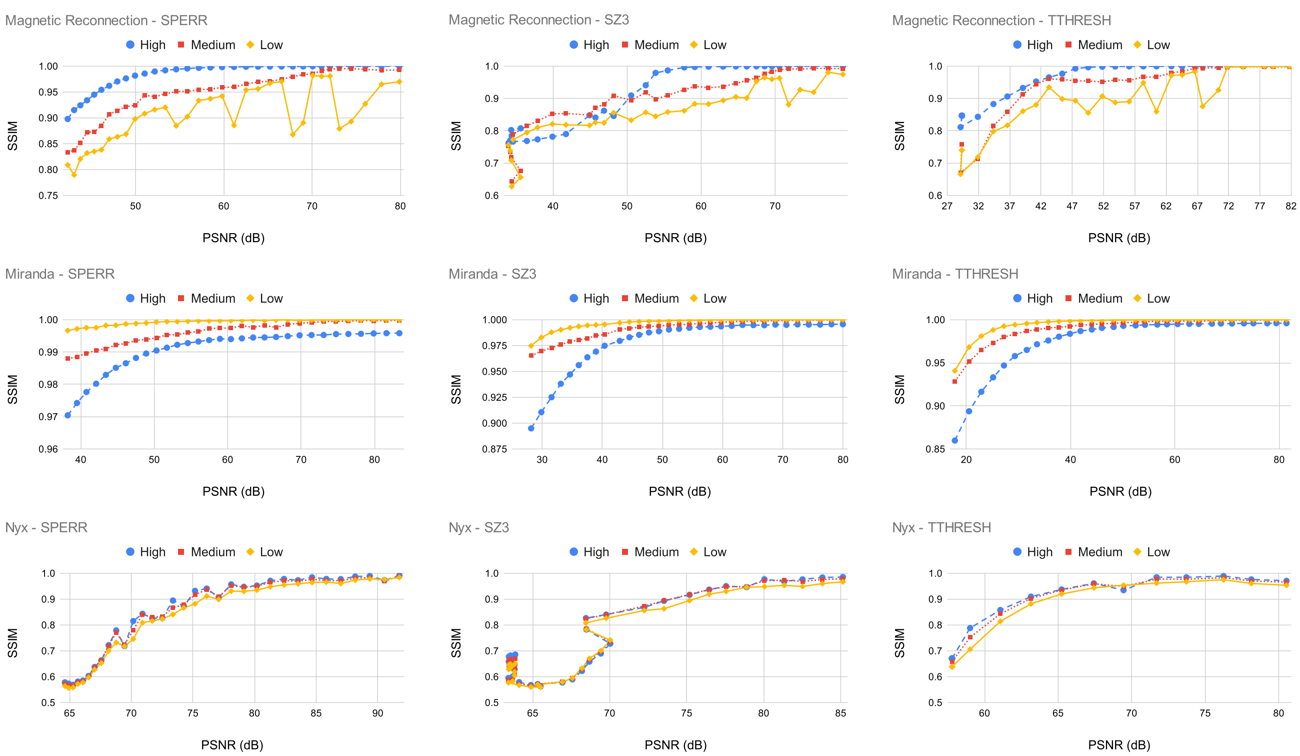


Fig. 8: The title of each graph specifies the dataset and compressor used for volume rendering each of the datasets with the red-blue color map at different opacities. Note that different scales are used for each graph.

we specified PSNR value to three compressors: SPERR, SZ3, and TTHRESH (in the range of 29 - 79); three datasets: Magnetic Reconnection, Miranda, and Nyx; and measured the resulting PSNR and compression ratio as shown in Figure 6. This *resulting PSNR* of the dataset is what we will use for our perception experiments. The PSNR range of 29 - 79 was chosen after an initial pilot study, with a small isotropic turbulence dataset [29], which showed that PSRN values within that range had the full spectrum from extremely poor perceptual quality to no perceptual difference for different opacities and colormaps.

The first observation is that different compressors produce different compression ratios for the same PSNR value. This is because compression algorithms use different underlying algorithms to reduce data size. The second observation is that, depending on the dataset being used, the compression ratio (CR) is quite different and the curves of the relationship of PSNR vs compression ratio can be quite different. This is accentuated in the case of the Nyx dataset: interestingly, when an initial PSNR value (PSNR error-bound) of 29 is specified for the Nyx dataset, the compressors returned a minimum PSNR value of 68 for SPERR, 64 for SZ3, and 57 for TTHRESH. We suspect that the very large value range of the Nyx dataset is artificially inflating the PSNR. Note that, the unsmooth PSNR vs. compression ratio curve for SZ3 on Nyx around CR=2500 (i.e., higher PSNR for higher Compression Ratios) occurs because, at very high compression ratios, SZ3's Huffman encoding may fail to select the optimal quantization bins, leading to performance fluctuations.

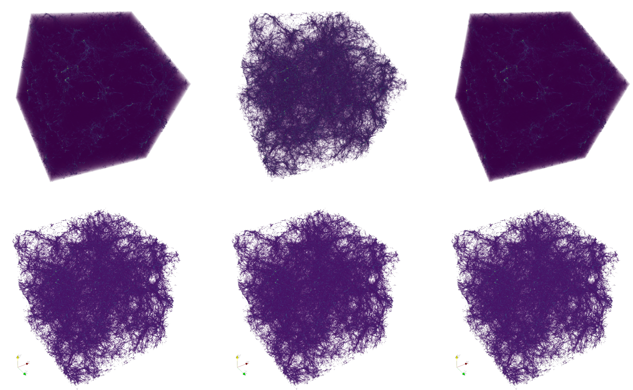
B. Impact of Color

One of the key elements of volume rendering is specifying a transfer function. A transfer function will specify the color as well as the opacity. Let us first take a look at colors; in other words, *does color impact error perception?* To do that, we investigate how the SSIM changes for different datasets and compressors per resulting PSNR, and show the result in Figure 7.

The first observation is that for the Miranda dataset, any data PSNR value of 40 and above has a very high SSIM value; all the graphs on the second row of Figure 7 (i.e. all the compressors) have SSIM values above 0.975 irrespective of the color used. The second observation is that color does not seem to have a major impact on the SSIM score. Apart from the Magnetic Reconnection dataset with SZ3, where the rainbow color map shows some wild variations, the same pattern of PSNR versus SSIM can be observed everywhere. The third observation is that the dataset being used has a huge impact on the quality. For the magnetic reconnection dataset, the PSNR has to be generally above 70, for Miranda, any PSNR above 40 will do, while for Nyx, we need to go as high as 80 for good quality.

C. Impact of Opacity

Opacity is crucial in volume rendering. As shown in Code Listing 1, opacity determines how far a ray travels in a volume:



PSNR 69 dB PSNR 73 dB PSNR 77 dB

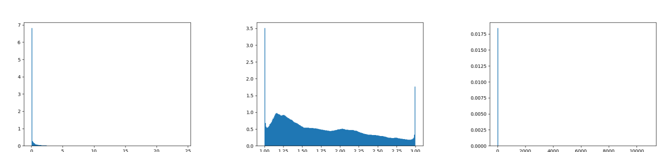
Fig. 9: Nyx dataset from three different PSNR volume rendered using the Viridis color map. In the top row, scalar value of 0 is included in the rendering while it is ignored in the bottom row. The same is observed using rainbow and red-blue color maps.

when opacity is low, a ray travels farther inside a volume and there is an increased probability that the ray will intersect voxels with larger errors. Figure 8 shows the results of varying the opacity for the red-blue color map.

The main observation is that the Magnetic Reconnection dataset has a quite different behavior from the two other datasets when it comes to its response to different opacities. We hypothesize that this is due to the anisotropic nature of the dataset and that huge regions of the dataset appear empty. These regions are not actually empty but rather they have regions of uniform data that we need to ignore; usually regions close to 0. To verify our hypothesis, we created two visualizations for the Nyx dataset where the scalar value 0 is ignored (bottom row of Figure 9) v/s 0 is ignored (top row of Figure 9). As can be seen, the variability close to 0 is again observed, and a PSNR value of 73 dB gives a more accurate visualization than a PSNR value of 77 dB.

D. Impact of Dataset

As discussed before and from Figures 7 and 8, the characteristics of a dataset have a major impact on how much a dataset can be compressed before the approximations introduced by a compressor are visibly perceptible; while a PSNR value of 40 might yield a very good image quality for the Miranda dataset, it is below acceptable for the Nyx dataset and barely



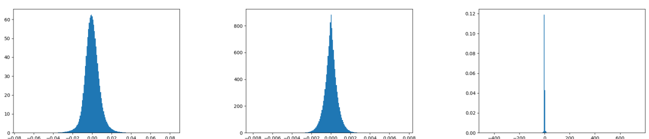
Magnetic Reconnection Miranda Nyx

Fig. 10: Histogram of data distribution; the x-axis shows scalar values while the y-axis shows density.

acceptable for the Magnetic Reconnection dataset. Why is that? Looking at Figure 10 and Table I, we can see that the Miranda dataset has a more uniform distribution than the Nyx and Magnetic Reconnection, and more importantly a very narrow range. At a PSNR value of ≈ 70, Table II and Figure 11, we can see very minor errors for the Miranda datasets compared to the other two, which would result in higher quality.

TABLE II: Statistics of the errors in the datasets.

Dataset	min	max	$\overline{x}$	σ
Magnetic Reconnection	-0.0732	0.0822	3.755e-06	0.00759
Miranda	-0.00794	0.00758	6.0843e-08	0.000649
Nyx	-461.355	737.692	-0.0360	3.3685

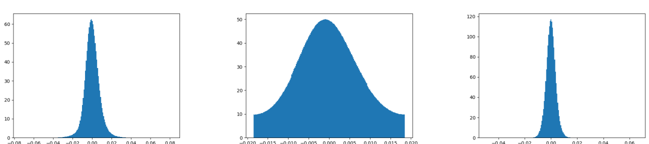


Magnetic Reconnection Miranda Nyx

Fig. 11: Histogram of error distribution (the x-axis shows the error distribution while the y-axis shows density) for the three test datasets at a PSNR score of 70 dB.

E. Impact of Compressors

Different compressors produce different results depending on the underlying algorithm used to perform data reduction. From Figures 7 and 8, we notice that TTHRESH and SPERR come close to an SSIM score of 1 faster than SZ3. We believe this is due to the distribution of errors, as seen in Figure 12, which is much narrower for SPERR and TTHRESH than SZ3.



SPERR SZ3 TTHRESH

Fig. 12: Histogram of error distribution (the x-axis shows the error distribution while the y-axis shows density) for the Magnetic Reconnection dataset PSNR at 70 dB for three different compressors.

V. Conclusion and Future Work

For this paper, we have conducted the first-ever rigorous study of the impact of lossy data reduction on volume rendering, analyzing the effect of opacity, color, dataset, and compressors. From our results, we observed that the most important factor that impacts perception quality is the scalar range of the dataset and the standard deviation of the error. While opacity can affect perception, color was interestingly not a significant factor.

A logical follow-up for this paper is to determine what is the impact on other visualization algorithms such as marching cubes [32], which is commonly used to extract iso-surfaces, slicing, and add lighting effects. Moreover, the same research protocol could be extended to include more data reduction algorithms, such as MGARD [33], SZ2, ZFP, as well as deep learning algorithms which are being increasingly researched for data reduction.

Moreover, in this work, we have seen that the PSNR of a dataset is an unreliable indicator of the quality of volume rendering visualization. To better understand the impact of data reduction on visualization, large scale user studies should be conducted to collect more data, which could be used to create better metrics that can predict visualization quality, for example, a more generic version of DSSIM [34] that could work for generic 3D data instead of only 2D data slices.

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APPENDIX

The graphs here show all the results per dataset for all the compressors (SPERR, SZ3 and TTHRESH), color maps (Rainbow, Red-Blue, and Viridis) and opacities (high, medium, and low).

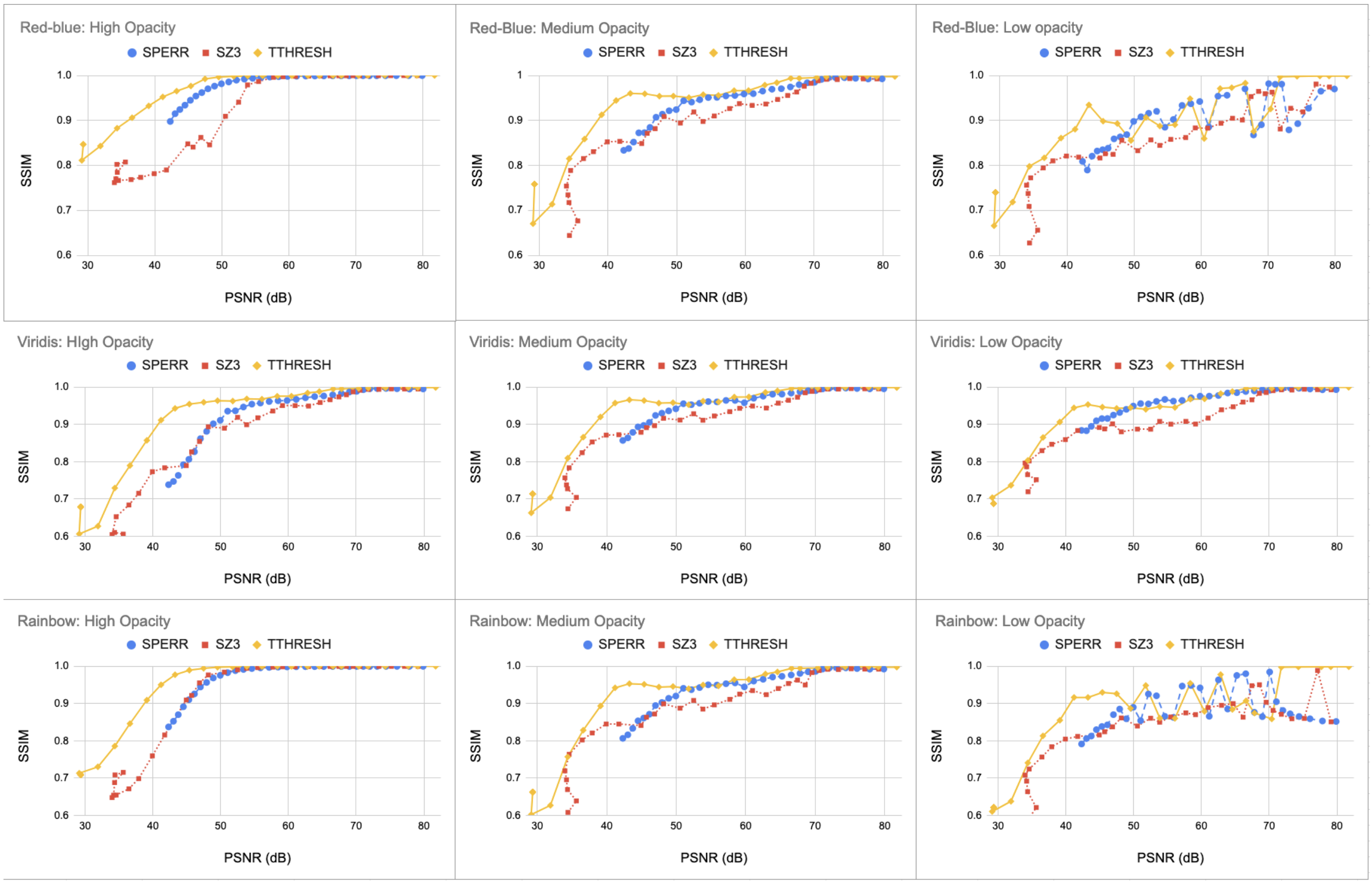

Fig. 13: Magnetic Reconnection Dataset

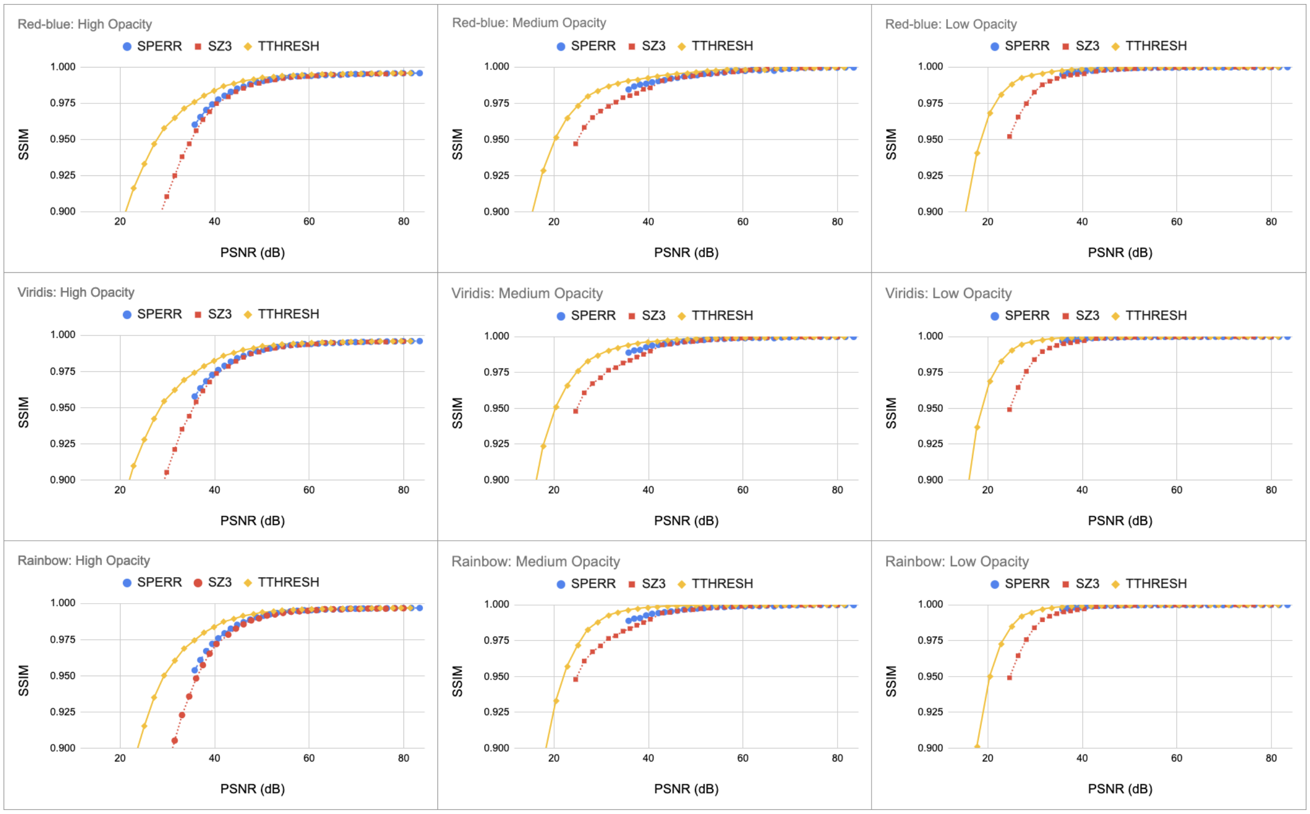

Fig. 14: Miranda Dataset

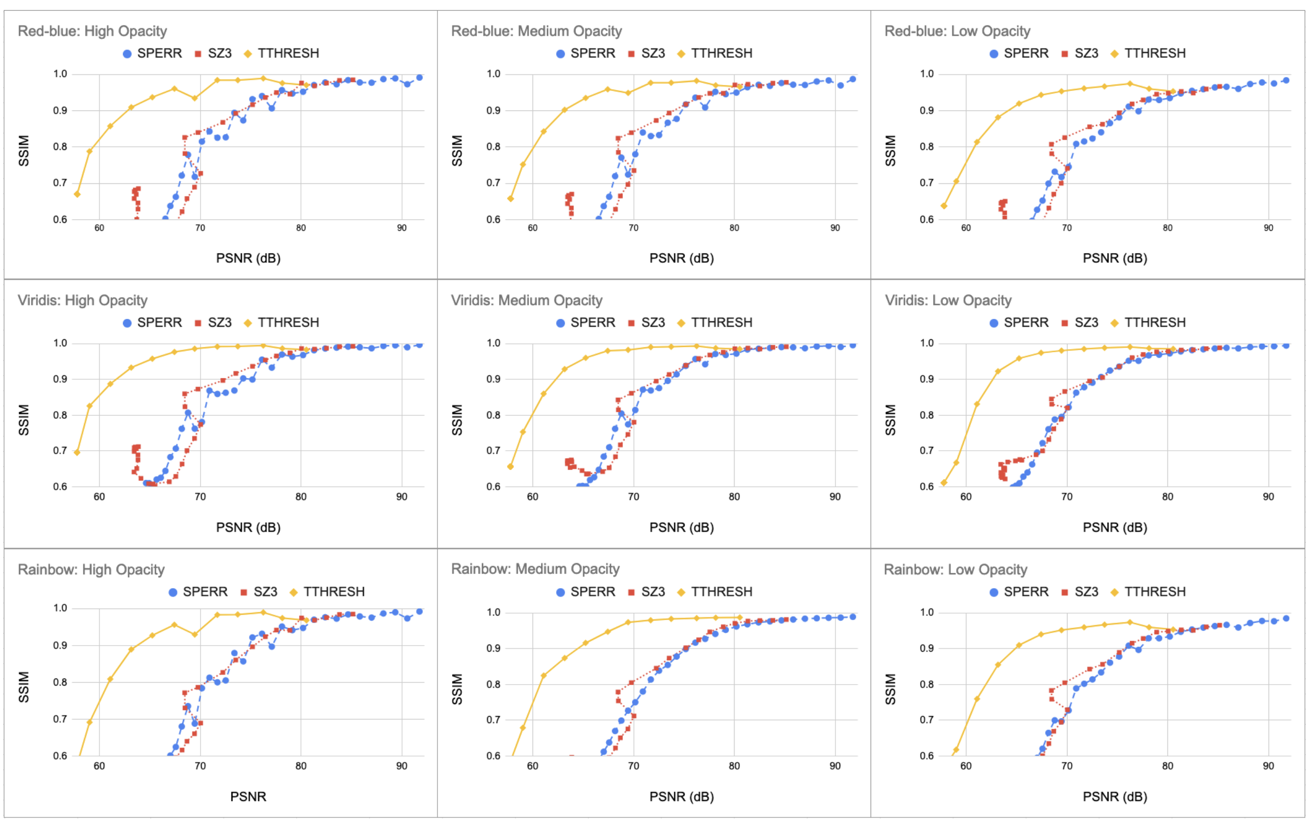

Fig. 15: Nyx Cosmology Dataset